CATEGORY B

Coding Assignment

Problem 3: Reverse the Queue

Difficulty: Medium · **Estimated time:** 25–30 minutes

TASK

At a ticket window, the queue order must be reversed so the last person is served first. Reverse the singly linked list in place.

Implement `Node reverse(Node head)` and return the new head. Write your own node class with `data` and `next` fields; do not use `java.util.LinkedList`.

INPUT / OUTPUT

Input: A list.

Output: The reversed list.

CONSTRAINTS

- 0 <= length of list <= 1000
- No new nodes may be created

EXAMPLES

Sample 1

```
queue = 1 -> 2 -> 3 -> 4

Expected Output: 4 -> 3 -> 2 -> 1 -> null
```

Sample 2

```
queue = 9

Expected Output: 9 -> null
```

COMPLEXITY REQUIREMENTS

State the time and additional space complexity.

HINT

Use three references — previous, current and next — and turn each next reference around one node at a time.